STPM 2025 PHYSICS 960/3

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ J s, $c = 3.00 \times 10^{8}$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $1\ u = 1.66 \times 10^{-27}$ kg, $N_A = 6.02 \times 10^{23}$ mol^{-1}.

Section A [15 marks]

1. A particle in an oscillatory system returns to an equilibrium position in the shortest time. Which is true about the oscillatory system?

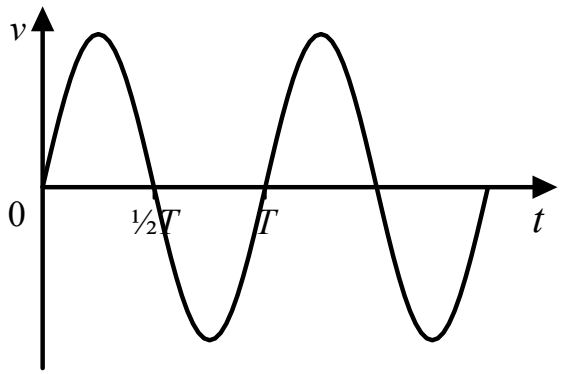
A An undamped system
B An underdamped system
C An overdamped system
D A critical damped system

Answer: D

A critically damped system returns to equilibrium in the shortest possible time without oscillating.

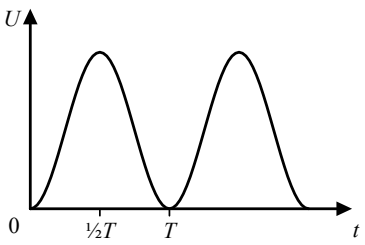
Explanation: An underdamped system overshoots and oscillates; an overdamped system returns slowly; an undamped system never settles.

2. The variation of velocity, v, with respect to time, t, of an object that undergoes a simple harmonic motion is shown in the graph below.

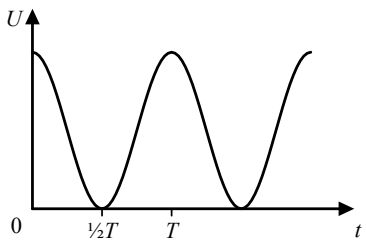


Which graph shows the variation of potential energy, U, with respect to time, t?

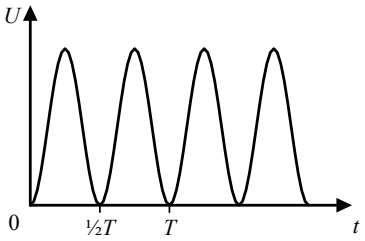
A



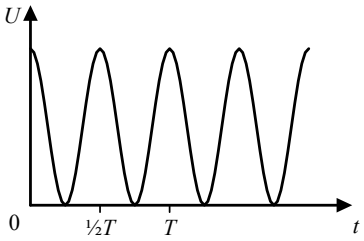
B



C



D



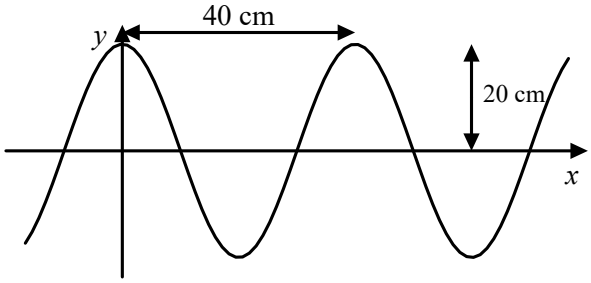
Answer: D

$v = v_0 \sin \omega t$, so at $t = 0$ the speed is zero and the object is at an extreme position: U is maximum.

$U \propto x^2$ repeats twice in each period, so U has period $\frac{1}{2}T$ and starts at its maximum (graph D).

Explanation: Potential energy is never negative and is maximum whenever the kinetic energy (velocity) is zero.

3. A progressive wave travels in positive-x direction at time, t, as shown in the graph below.



If the frequency of the wave is 8.0 Hz, which expression represents the above progressive wave?

A $y = 0.20 \sin (16\pi t - 5\pi x)$
B $y = 0.20 \sin (16\pi t + 5\pi x)$
C $y = 0.40 \sin (16\pi t - 5\pi x)$
D $y = 0.40 \sin (16\pi t + 5\pi x)$

Answer: A

$$\text{Amplitude} = 20 \text{ cm} = 0.20 \text{ m}; \lambda = 40 \text{ cm} = 0.40 \text{ m} \Longrightarrow k = \frac{2\pi}{0.40} = 5\pi \text{ m}^{-1}$$

$\omega = 2\pi(8.0) = 16\pi$ rad s^{-1}; travelling in $+x$: $y = 0.20 \sin (16\pi t - 5\pi x)$

Explanation: A wave moving in the positive x-direction has opposite signs in front of ωt and kx. The 40 cm on the graph is between two crests, i.e. one wavelength; the 20 cm is the amplitude.

4. Two sinusoidal progressive waves superpose to form a resultant progressive wave. The resultant progressive wave is also sinusoidal if

A the progressive waves have different amplitude and travel in the same direction
B the progressive waves have the same frequency and travel in the same direction
C the progressive waves have different amplitude but travel in the opposite direction
D the progressive waves have the same frequency but travel in the opposite direction

Answer: B

Two sinusoidal waves of the same frequency travelling in the same direction add to give another sinusoidal progressive wave of that frequency.

Explanation: Waves of the same frequency travelling in opposite directions form a stationary wave, not a progressive one.

5. A stretched string is fixed at its two ends. When the string is plucked at a point, the string oscillates in the second overtone frequency. How many antinodes are present on the string?

A One B Two C Three D Four

Answer: C

Second overtone = third harmonic: the string vibrates in 3 loops, so there are 3 antinodes (and 4 nodes).

Explanation: For a string fixed at both ends, the *n*th harmonic has *n* loops (antinodes). The first overtone is the second harmonic.

6. What is the fundamental frequency of a tube that is closed at one of its end if the length of the tube is 1.20 m?

[Speed of sound in air is 346 m s^{-1}.]

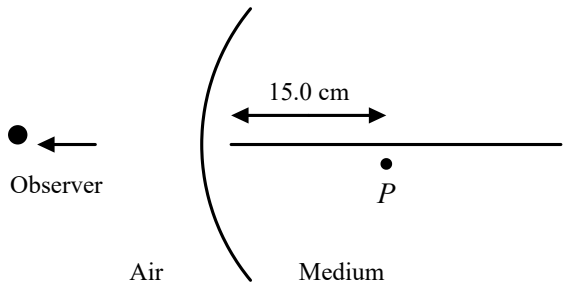
A 72.1 Hz B 144 Hz C 216 Hz D 288 Hz

Answer: A

$$f_0 = \frac{v}{4L} = \frac{346}{4(1.20)} = 72.1 \text{ Hz}$$

Explanation: A tube closed at one end has a node at the closed end and an antinode at the open end: $L = \lambda/4$ for the fundamental.

7. A portion of spherical surface which has a refractive index of 1.40 is placed in air and an observer are shown as in the diagram below.



An object, *P*, is placed in a medium 15.0 cm from the curved surface. If the radius of curvature is 20.0 cm, what is the distance of the image of *P* from the curved surface as seen by the observer?

A 8.82 cm in the air
B 8.82 cm in the medium
C 13.6 cm in the air
D 13.6 cm in the medium

Answer: D

$$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r} \text{ with } n_1 = 1.40,\ n_2 = 1.00,\ u = 15.0 \text{ cm},\ r = -20.0 \text{ cm}$$

$$\frac{1.40}{15.0} + \frac{1}{v} = \frac{1.00 - 1.40}{-20.0} \Rightarrow v = -13.6 \text{ cm}$$

Negative v: a virtual image 13.6 cm from the surface, in the medium.

Explanation: Light travels from the medium into the air. The surface bulges towards the air, so the centre of curvature is on the medium (incident) side and r is negative in the real-is-positive convention.

8. The radius of curvature of a plano-convex lens is 20.0 cm and its refractive index is 1.73. What is the power of the lens?

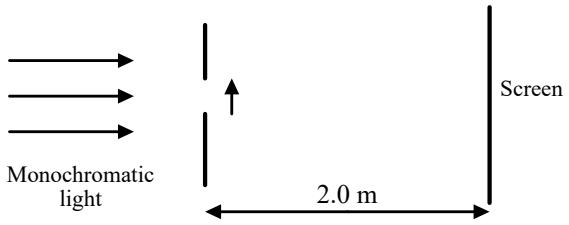
A 8.65 D B 3.65 D C 0.087 D D 0.036 D

Answer: B

$$P = \frac{1}{f} = (n - 1)\frac{1}{r} = \frac{0.73}{0.200} = 3.65 \text{ D}$$

Explanation: For a plano-convex lens the flat face has infinite radius, so only one term remains in the lensmaker's equation. The radius must be in metres.

9. The setup of Young's two-slits experiment which has a separation of 0.50 mm is shown in the diagram below.



Interference patterns observed on the screen consists of dark and bright fringes. If the separation between two adjacent bright fringes is 2.4 mm, what is the wavelength of the monochromatic light used?

A 400 nm B 500 nm C 600 nm D 700 nm

Answer: C

$$\lambda = \frac{yd}{D} = \frac{(2.4 \times 10^{-3})(0.50 \times 10^{-3})}{2.0} = 6.0 \times 10^{-7} \text{ m} = 600 \text{ nm}$$

Explanation: The fringe separation is $y = \lambda D/d$, where D is the slit–screen distance (2.0 m).

10. Which is true about fibre optics?

A Heavier than metallic conductors
B Affected by electromagnetic interference
C Emission of radiation to the surrounding does not occur
D Transmission of light for long distance is due to long and thick strands of glass

Answer: C
Light is kept inside the fibre by total internal reflection, so (almost) no radiation is emitted to the surroundings.

Explanation: Optical fibres are light, immune to electromagnetic interference and are made of thin strands of glass.

11. A bullet of mass 0.020 kg is shot from a rifle. If the kinetic energy of the bullet is 900 J, what is the de Broglie wavelength of the bullet?

A 3.7×10^{-17} m B 2.2×10^{-36} m C 1.1×10^{-34} m D 3.3×10^{-32} m

Answer: C

$$p = \sqrt{(2mK)} = \sqrt{(2(0.020)(900))} = 6.0 \text{ kg m s}^{-1}$$

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{6.0} = 1.1 \times 10^{-34} \text{ m}$$

Explanation: The wavelength is far too small to observe, which is why everyday objects show no wave behaviour.

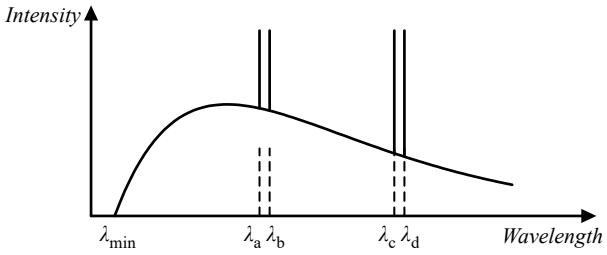
12. Which statement is **not** true about nanoscience?

A Nanoscience is the study of structures of size less than 100 nanometres.
B Nanotechnology can be used to reduce the size of transistors in integrated circuits.
C Nanosized particles exhibit similar properties as the larger particles of the same substance.
D Nanotransistors can increase the processing speed and memory of microprocessors in computers significantly.

Answer: C
Nanosized particles often have properties very different from larger particles of the same substance (because of their large surface area to volume ratio and quantum effects). Statement C is not true.

Explanation: This difference in properties is exactly what makes nanotechnology useful.

13. The X-ray spectrum produced by a metal target in an X-ray tube is shown in the diagram below.



Which deduction can be made about the X-rays produced by the metal target?

A All metals produce characteristic line spectrum at the same wavelengths.
B The X-ray photons with a minimum wavelength emitted have the lowest energy.
C The X-ray photons with the wavelengths in the characteristic line spectrum emitted have the highest energy.
D The various wavelengths of X-ray photons in the continuous spectrum is due to the deceleration of electrons when the electrons hit the metal target.

Answer: D

The continuous spectrum (bremsstrahlung) is produced when electrons decelerate as they hit the target, losing different amounts of energy, so a range of wavelengths is emitted.

Explanation: Characteristic lines depend on the target metal (A wrong); the minimum wavelength corresponds to the **highest** photon energy (B wrong); the line photons are not the most energetic (C wrong).

14. Which statement is **not** true about mass-energy relationship?

A Energy can be converted to mass.
B Mass can be converted to energy.
C Mass is always conserved in a nuclear reaction.
D Energy, E, which is equivalent to the mass, m, is given by $E = mc^2$.

Answer: C

In a nuclear reaction mass is not conserved on its own: some mass is converted to energy ($E = mc^2$). It is the total mass–energy that is conserved.

Explanation: The mass defect of the products is what releases energy in fission, fusion and radioactive decay.

15. The nucleus of radium undergoes alpha decay according to the equation $^{226}_{88}\text{Ra} \rightarrow {}^{222}_{86}\text{Rn} + {}^{4}_{2}\text{He}$. What is the energy released during the decay?

[Mass of $^{226}_{88}\text{Ra}$ is 226.0254 u, mass of $^{222}_{86}\text{Rn}$ is 222.0176 u and mass of $^{4}_{2}\text{He}$ is 4.0026 u.]

A 6.8×10^{8} J B 7.8×10^{-13} J C 2.6×10^{-21} J D 8.7×10^{-30} J

Answer: B

$$\Delta m = 226.0254 - (222.0176 + 4.0026) = 0.0052 \text{ u}$$

$$E = \Delta mc^2 = (0.0052)(1.66 \times 10^{-27})(3.00 \times 10^8)^2 = 7.8 \times 10^{-13} \text{ J}$$

Explanation: Convert the mass defect from u to kg before using $E = mc^2$. (1 u ≈ 931 MeV gives 4.8 MeV, the same energy.)

Section B [15 marks]

16. A small block is attached to a horizontal spring and then is compressed and released. The block then oscillates on a frictionless surface with a period of 1.2 s and an amplitude of 0.060 m.

(a) Write an equation of the displacement for the block with time t after being released. [3]

Answer

$\omega = \frac{2\pi}{T} = \frac{2\pi}{1.2} = 5.24 \text{ rad s}^{-1}$ — 1

Released from maximum compression: $x = -A \cos \omega t$ — 1

$x = -0.060 \cos (5.24t)$ m — 1

3

Explanation: The block starts at an extreme position (taken as −0.060 m, compressed), so a cosine function fits. Sine would start at the equilibrium position.

(b) Determine the velocity of the block at time 1.6 s. [3]

Answer

$v = \frac{dx}{dt} = A\omega \sin \omega t$ — 1

$v = (0.060)(5.24) \sin (5.24 \times 1.6) = 0.314 \sin (8.38 \text{ rad})$ — 1

$v = 0.272 \text{ m s}^{-1}$ — 1

3

Explanation: Use radians when finding the sine. The positive value shows the block is moving in the positive (extension) direction at that moment.

(c) Determine the maximum acceleration of the block. [2]

Answer

$a_{max} = A\omega^2 = (0.060)(5.24)^2$ — 1

$a_{max} = 1.64 \text{ m s}^{-2}$ — 1

2

Total for question **8**

Explanation: The acceleration is greatest at the extreme positions, where the displacement is greatest.

17. Electrons are ejected from the surface of a silver coated metal plate when exposed to electromagnetic radiation. The work function of silver is 4.26 eV.

(a) Determine the longest wavelength of electromagnetic radiation that can eject an electron from the surface of the plate. [3]

Answer

Threshold: $W = \frac{hc}{\lambda_0} \Longrightarrow \lambda_0 = \frac{hc}{W}$ 1

$\lambda_0 = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{(4.26)(1.60 \times 10^{-19})}$ 1

$\lambda_0 = 2.92 \times 10^{-7}$ m 1

3

Explanation: Convert the work function from eV to joules. Longer wavelengths have photons with too little energy to release electrons.

(b) If the metal plate is exposed to monochromatic light of wavelength 200 nm, determine

(i) the maximum kinetic energy of the photoelectrons, [2]

Answer

$K_{max} = \frac{hc}{\lambda} - W = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{200 \times 10^{-9}} - 6.82 \times 10^{-19}$ 1

$K_{max} = 3.13 \times 10^{-19}$ J (1.96 eV) 1

2

Explanation: Einstein's photoelectric equation: photon energy = work function + maximum kinetic energy.

(ii) the stopping potential of the emitted electrons. [2]

Answer

$eV_s = K_{max} \Longrightarrow V_s = \frac{3.13 \times 10^{-19}}{1.60 \times 10^{-19}}$ 1

$V_s = 1.96$ V 1

2

Total for question **7**

Explanation: The stopping potential just stops the fastest electrons; numerically it equals K_{max} in eV.

Section C [30 marks]

18. (a) State **one** factor that affects the intensity of sound. [1]

Answer

Any one: amplitude of the sound / power of the source / distance from the source — 1

1

Explanation: Intensity ∝ amplitude^2, and for a point source $I = P/4\pi r^2$.

(b) The average intensity level for a violin is 65 dB. Determine the average intensity level when 20 similar violins are played simultaneously. [3]

Answer

Intensity becomes $20I$: $\beta = 10 \log \frac{20I}{I_0} = 65 + 10 \log 20$ — 1

$= 65 + 13.0$ — 1

$= 78.0$ dB — 1

3

Explanation: Intensities add, not decibel levels. Multiplying the intensity by 20 adds 10 log 20 = 13.0 dB.

(c) An intensity level of a sound at 4.0 m from a small aircraft is 100 dB. Determine the intensity level at the position of 10 m from the aircraft.

[Threshold intensity for hearing is 1.0×10^{-12} W m^{-2}.] [3]

Answer

$I \propto \frac{1}{r^2} \Rightarrow \frac{I_2}{I_1} = (\frac{4.0}{10})^2 = 0.16$ — 1

$\beta_2 = 100 + 10 \log 0.16 = 100 - 7.96$ — 1

$\beta_2 = 92.0$ dB — 1

3

Explanation: Treat the aircraft as a point source. (Equivalently, find $I_1 = 1.0 \times 10^{-2}$ W m^{-2}, then $I_2 = 1.6 \times 10^{-3}$ W m^{-2}.)

(d) Two submarines, P and Q, are moving towards each other. P and Q move at the speed of 15.0 m s^{-1} and 5.0 m s^{-1} respectively. P sends a sonar signal of frequency 10.0 kHz to Q. The speed of sound in water is 1500 m s^{-1}.

(i) Calculate the apparent frequency detected by Q. [2]

Answer

Both approaching: $f' = f\frac{v + v_o}{v - v_s} = (10.0 \times 10^3)\frac{1500 + 5.0}{1500 - 15.0}$ — 1

$f' = 1.01 \times 10^4$ Hz (10 135 Hz) — 1

2

Explanation: P is the source moving at 15.0 m s^{-1}; Q is the observer moving at 5.0 m s^{-1}. Both motions raise the frequency.

(ii) Why does apparent frequency detected by Q different from the actual frequency? Explain your answer. [3]

Answer

There is relative motion between the source and the observer (Doppler effect) — 1

The moving source P catches up with its own waves, so the wavelength in front of it is shorter — 1

The moving observer Q meets more wavefronts per second; both effects make the apparent frequency higher than 10.0 kHz — 1

3

Explanation: The speed of sound in water does not change; only the wavelength (source motion) and the rate at which waves are met (observer motion) change.

(iii) Calculate the frequency of echo received by P from Q. [3]

Answer

Q reflects the signal and acts as a source of frequency 10 135 Hz moving at 5.0 m s^{-1} towards P, which moves at 15.0 m s^{-1} — 1

$f'' = (10\ 135)\frac{1500 + 15.0}{1500 - 5.0}$ — 1

$f'' = 1.03 \times 10^4$ Hz (10 271 Hz) — 1

3

Total for question **15**

Explanation: The echo undergoes a second Doppler shift, with the roles of source and observer exchanged.

19. (a) Explain the process of radioactive decay. [4]

Answer

Unstable nuclei emit α particles, β particles and/or γ rays to become more stable	1
The process is spontaneous: it is not affected by external conditions such as temperature or pressure	1
The process is random: it is impossible to predict which nucleus will decay next	1
The rate of decay is proportional to the number of undecayed nuclei, $dN/dt = -\lambda N$; a new nuclide is formed (except in γ emission)	1
	4

Explanation: Randomness and spontaneity are the two key words; the exponential decay law follows from them.

(b) A patient is injected by 0.010 μg of technetium, Tc-99m, for medical diagnosis. After 2 hours, 20% of Tc-99m has decayed.

(i) Determine the remaining Tc-99m after 5 hours. [3]

Answer

80% remains every 2 hours: $m = m_0(0.8)^{t/2}$	1
$m = (0.010)(0.8)^{2.5}$	1
$m = 5.72 \times 10^{-3}$ μg	1
	3

Explanation: Exponential decay: the same fraction decays in each equal time interval. (Using λ from (b)(ii) gives the same result.)

(ii) Determine the activity of Tc-99m after 5 hours. [3]

Answer

$0.8 = e^{-\lambda(7200)} \Longrightarrow \lambda = \frac{\ln 1.25}{7200} = 3.10 \times 10^{-5}\ s^{-1}$	1
$N = \frac{5.72 \times 10^{-9}}{98.9} \times 6.02 \times 10^{23} = 3.48 \times 10^{13}$	1
$A = \lambda N = (3.10 \times 10^{-5})(3.48 \times 10^{13}) = 1.08 \times 10^{9}$ Bq	1
	3

Explanation: Use the decay constant in s^{-1} so that the activity is in becquerels. Convert micrograms to grams before dividing by the molar mass.

(iii) Calculate the half-life of Tc-99m.

[Average atomic mass of Tc-99m is 98.9 g mol^{-1}.] [2]

Answer

$T_{½} = \frac{\ln 2}{\lambda} = \frac{0.693}{3.10 \times 10^{-5}} = 2.24 \times 10^{4}$ s — 1

$T_{½} = 6.21$ h — 1

2

Explanation: Tc-99m's half-life of about 6 hours is short enough to limit the patient's dose but long enough for the diagnosis.

(c) State **three** differences between fission and fusion processes. [3]

Answer

• Fission splits a heavy nucleus into two lighter nuclei; fusion joins two light nuclei into a heavier nucleus — 1

• Fission can occur at ordinary temperatures (started by a neutron); fusion needs extremely high temperature and pressure — 1

• Fusion releases more energy per unit mass (per nucleon); fission produces radioactive waste while fusion produces little / fission can be a chain reaction — 1

3

Total for question **15**

Explanation: In both processes the binding energy per nucleon of the products is greater than that of the reactants, so energy is released.

20. (a)

(i) State the Huygen's principle. [3]

Answer

Every point on a wavefront acts as a source of secondary wavelets — 1

The secondary wavelets spread out in the forward direction with the same speed (and frequency) as the original wave — 1

The new wavefront is the surface tangent to (the envelope of) all the secondary wavelets — 1

3

Explanation: Huygens' principle explains reflection, refraction and diffraction by constructing new wavefronts.

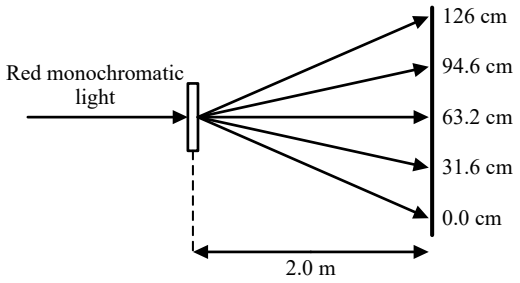
(ii) Explain the diffraction pattern for a single slit. [4]

Answer

Each point in the slit acts as a source of secondary wavelets, which interfere on the screen	1
Bright central maximum, twice as wide as the other maxima and most intense	1
Minima where $a \sin\theta = m\lambda$ ($m = 1, 2, \ldots$), where wavelets from the two halves of the slit cancel in pairs	1
Secondary maxima of much lower and decreasing intensity on both sides	1
	4

Explanation: The narrower the slit (compared with the wavelength), the wider the central maximum.

(b) A diffraction grating of 3.0×10^5 lines per metre is placed in front of a source of red monochromatic light and the diffraction pattern is produced on a screen as shown in the diagram below.



The screen is placed 2.0 m from the grating. The distance between diffraction pattern of the bright fringes are in the range from 0.0 cm to 126 cm.

(i) Based on the first order maximum, determine the wavelength of the light. [5]

Answer

Central (zero-order) maximum is at 63.2 cm (straight through the grating)	1
First-order maximum is $63.2 - 31.6 = 31.6$ cm from the centre	1
$\tan\theta = \dfrac{0.316}{2.0} \Longrightarrow \theta = 8.98°$	1
$d = \dfrac{1}{3.0 \times 10^5} = 3.33 \times 10^{-6}$ m	1
$\lambda = d \sin\theta = (3.33 \times 10^{-6}) \sin 8.98° = 5.2 \times 10^{-7}$ m	1
	5

Explanation: The pattern is symmetrical about the central maximum, so measure distances from 63.2 cm. Grating equation: $d \sin\theta = n\lambda$ with $n = 1$.

(ii) Suggest advantage and disadvantage of using second order maximum to calculate the wavelength of the light. [2]

Answer

Advantage: the angle is larger, so the percentage error in measuring the distance/angle is smaller	1
Disadvantage: the second-order maximum is less intense (dimmer), so its position is harder to locate	1
	2

Explanation: Higher orders are more spread out but fainter.

(iii) State the change of diffraction pattern if the source is replaced by blue monochromatic light. [1]

Answer

The bright fringes move closer together (smaller angles), because blue light has a shorter wavelength	1
	1
Total for question	**15**

Explanation: $\sin\theta = n\lambda/d$: a smaller wavelength gives a smaller diffraction angle for each order.